

Angle Measures on Parallel Lines

Answer Key

High-School Geometry Lines and Angles

Quick Answers

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|------------|------------|------------|-------------|
| 1. 62, 118 | 4. 125, 55 | 7. 16, 76 | 10. 14, 122 |
| 2. 133, 47 | 5. 12, 69 | 8. 22, 102 | |
| 3. 17, 61 | 6. 108, 72 | 9. 15, 64 | |
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Curriculum

Level: High-School Geometry Lines and Angles

HSG-CO.C.9 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 21 tagged checks.

Common wrong answers

3. *If they got 17: stopped at x and reported it as the angle measure*
7. *If they got 20: set the two expressions equal instead of adding them to 180, getting x of 2 and an angle of 20*
8. *If they got 22: stopped at x and reported it as the angle measure*
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Angle names refer to the shared diagram on the worksheet: $\ell \parallel m$ cut by transversal t , with a, b, c, d around the upper intersection and e, f, g, h around the lower one.

1. $m\angle a = 118^\circ$. Find $m\angle d$ and $m\angle e$.

Step 1. $\angle a$ and $\angle d$ sit on the same side of the transversal on either side of line ℓ , so together they make a straight angle — a linear pair.

Step 2. $180 - 118 = 62$, so $m\angle d$ is

62°

Step 3. $\angle e$ occupies the same position at line m that $\angle a$ occupies at line ℓ , so they are corresponding angles, and the lines are parallel — corresponding angles are therefore congruent.

118°

Listen for: “they look about the same”. The diagram is not to scale; the reason is the parallel-line postulate, not the picture.

2. $m\angle c = 47^\circ$. Find $m\angle f$ and $m\angle g$.

Step 1. $\angle c$ and $\angle f$ are both between the two parallel lines and both on the same side of the transversal — same-side interior angles, which are supplementary.

Step 2. $180 - 47 = 133$.

133°

Step 3. $\angle g$ corresponds to $\angle c$, so the two are congruent.

47°

Teaching note: same-side interior is the one pair on this diagram that adds to a straight angle rather than matching. Students who memorise “parallel means equal” get this one wrong every time.

3. $m\angle b = (3x + 10)^\circ$, $m\angle f = (5x - 24)^\circ$.

Step 1. $\angle b$ and $\angle f$ are corresponding angles, so with $\ell \parallel m$ their measures are equal. Set the expressions equal.

Step 2. $3x + 10 = 5x - 24$ gives $34 = 2x$, so

$$x = 17$$

Step 3. Substitute back into either expression: $3(17) + 10 = 51 + 10$.

$$61^\circ$$

Common wrong answer: 17 given as the angle. Finding x is only the first half; the question asked for a measure in degrees. Checking the other expression is worthwhile too: $5(17) - 24 = 61$ as well.

4. $m\angle h = 125^\circ$. Find $m\angle b$ and $m\angle e$.

Step 1. $\angle b$ and $\angle h$ are both outside the parallel lines and on opposite sides of the transversal — alternate exterior angles, which are congruent.

$$125^\circ$$

Step 2. $\angle h$ and $\angle e$ are adjacent along line m on the same side of it, so they form a linear pair.

Step 3. $180 - 125 = 55$.

$$55^\circ$$

Teaching note: alternate exterior is the pair students meet least often. Point out that it is the mirror image of alternate interior, taken outside the parallel lines instead of between them.

5. $m\angle d = (7x - 15)^\circ$, $m\angle f = (4x + 21)^\circ$.

Step 1. Both angles lie between the parallel lines and on opposite sides of the transversal — alternate interior angles, so they are congruent.

Step 2. $7x - 15 = 4x + 21$ gives $3x = 36$, so

$$x = 12$$

Step 3. $7(12) - 15 = 84 - 15$.

$$69^\circ$$

Check it: the other expression gives $4(12) + 21 = 69$ as well, which is the fastest way for a student to catch an arithmetic slip.

6. $m\angle f = 108^\circ$. Find $m\angle d$ and $m\angle a$.

Step 1. $\angle d$ and $\angle f$ are alternate interior angles, so they are congruent.

$$108^\circ$$

Step 2. There is no single named pair joining $\angle f$ to $\angle a$, so go through $\angle d$: $\angle a$ and $\angle d$ form a linear pair at the upper intersection.

Step 3. $180 - 108 = 72$.

$$72^\circ$$

Teaching note: this is the first problem that needs two relationships chained together. Writing the middle result down before reaching for the second rule is the habit worth building.

7. $m\angle c = (4x + 12)^\circ$, $m\angle f = (6x + 8)^\circ$.

Step 1. $\angle c$ and $\angle f$ are same-side interior angles, so they are supplementary, not congruent. Their sum is 180.

Step 2. $(4x + 12) + (6x + 8) = 180$ gives $10x + 20 = 180$, then $10x = 160$, so

$$x = 16$$

Step 3. $4(16) + 12 = 64 + 12$.

$$76^\circ$$

Common wrong answer: 20 — from setting the two expressions equal, which gives $x = 2$. That is the congruent rule applied to a supplementary pair. Sanity check: $76 + 104 = 180$, as a same-side interior pair must.

8. $m\angle d = (5x - 8)^\circ$, $m\angle e = (3x + 12)^\circ$.

Step 1. $\angle d$ is interior at the upper line and $\angle e$ is interior at the lower line, both on the same side of the transversal — so they are supplementary.

Step 2. $(5x - 8) + (3x + 12) = 180$ gives $8x + 4 = 180$, then $8x = 176$, so

$$x = 22$$

Step 3. $5(22) - 8 = 110 - 8$.

$$102^\circ$$

Common wrong answer: 22 reported as the angle. Check the pair: $m\angle e = 3(22) + 12 = 78$, and $102 + 78 = 180$.

9. $m\angle a = (2x + 34)^\circ$, $m\angle g = (5x - 11)^\circ$.

Step 1. $\angle a$ is outside the parallel lines at the upper intersection and $\angle g$ is outside them at the lower one, on the other side of the transversal — alternate exterior angles, so congruent.

Step 2. $2x + 34 = 5x - 11$ gives $45 = 3x$, so

$$x = 15$$

Step 3. $2(15) + 34 = 30 + 34$.

$$64^\circ$$

Teaching note: a student who cannot see the pair can still get there in two steps — $\angle a$ and $\angle c$ are vertical angles, and $\angle c$ and $\angle g$ correspond.

10. $m\angle b = (9x - 4)^\circ$, $m\angle e = (3x + 16)^\circ$ — **decide the relationship yourself.**

Step 1. $\angle b$ and $\angle e$ are not one of the named pairs, so chain two that are: $\angle b$ and $\angle f$ correspond, hence are congruent, and $\angle f$ and $\angle e$ form a linear pair. So $\angle b$ and $\angle e$ are supplementary — they add to a straight angle, they are not equal.

Step 2. $(9x - 4) + (3x + 16) = 180$ gives $12x + 12 = 180$, then $12x = 168$, so

$$x = 14$$

Step 3. $9(14) - 4 = 126 - 4$.

$$122^\circ$$

Step 4. Check: $m\angle e = 3(14) + 16 = 58$, and $122 + 58 = 180$.

Open response (manual review): the written justification is the student's own. Accept any answer that reaches supplementary through a valid chain — corresponding then linear pair, as above, or vertical angles then same-side interior. Do not accept “they looked supplementary”; the point of the question is the chain of reasons. A student who assumed congruent gets $6x = 20$ and a non-integer x , which is itself a signal that the wrong rule was chosen.